Ratio, Root, and Alternating Series Tests

ANSWER KEY

Section 1: The Ratio Test

1 Evaluate the limit $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for the series $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

a. 0 **(converges)** ✓

b. 3 (diverges)

c. 1 (inconclusive)

d. ∞ (diverges)

$\lim_{n\to\infty} \frac{3^{n+1}}{(n+1)!}\frac{n!}{3^n} = \lim_{n\to\infty} \frac{3}{n+1} = 0 < 1$. By the Ratio Test, the series converges.

2 When is the Ratio Test inconclusive?

a. **When** $L = 1$ ✓

b. When $L < 1$

c. When $L > 1$

d. When $L = 0$

The Ratio Test is inconclusive whenever the limit $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1$, requiring another test (such as comparison or integral test).

3 Determine the convergence of $\sum_{n=1}^{\infty} \frac{n!}{2^n}$.

a. **Diverges because** $L = \infty > 1$ ✓

b. Converges because $L = 0 < 1$

c. Inconclusive ($L = 1$)

d. Converges by Alternating Series Test

$\lim_{n\to\infty} \frac{(n+1)!}{2^{n+1}}\frac{2^n}{n!} = \lim_{n\to\infty} \frac{n+1}{2} = \infty > 1$. The series diverges.

Section 2: The Alternating Series Test

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4 What two conditions must an alternating series $\sum(-1)^{n-1}b_n$ with $b_n > 0$ satisfy to converge by the AST?

a. $b_{n+1} \le b_n$ **for all** n**, and** $\lim_{n\to\infty} b_n = 0$ ✓

b. b_n is increasing, and $\lim_{n\to\infty} b_n = 1$

c. $\lim_{n\to\infty} b_n = 0$ only

d. The Ratio Test limit must be < 1

By the Alternating Series Test (Leibniz's rule), the terms must be non-increasing in magnitude ($b_{n+1} \le b_n$) and must approach zero ($\lim_{n\to\infty} b_n = 0$).

5 Does the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converge or diverge?

a. **Converges conditionally** ✓

b. Converges absolutely

c. Diverges

d. Oscillates without limit

It converges by the AST ($1/n$ decreases to 0), but $\sum|a_n| = \sum\frac{1}{n}$ diverges (harmonic series). Thus, it converges conditionally.

6 Determine the convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$.

a. **Converges conditionally** ✓

b. Converges absolutely

c. Diverges

d. Inconclusive

Terms decrease to 0, so it converges by the AST. The absolute series $\sum n^{-1/2}$ is a divergent p-series ($p = 0.5 \le 1$). Hence conditional convergence.

Section 3: Absolute vs. Conditional Convergence

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7 A series $\sum a_n$ is said to converge absolutely if which condition holds?

a. $\sum |a_n|$ **converges** ✓

b. $\sum a_n$ converges but $\sum |a_n|$ diverges

c. $\lim_{n\to\infty} a_n = 0$

d. All terms are positive

Absolute convergence is defined by the convergence of the series of absolute values: $\sum |a_n|$ converges.

8 Which of the following series converges absolutely?

a. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ ✓

b. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

c. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} (-1)^n$

The absolute series is $\sum \frac{1}{n^2}$, which is a convergent p-series ($p = 2 > 1$). Therefore, it converges absolutely.

9 If a series converges absolutely, MUST it also converge in its original form?

a. **Yes, absolute convergence implies convergence** ✓

b. No, it may oscillate

c. Only if it is alternating

d. Only for geometric series

A fundamental theorem of analysis guarantees that every absolutely convergent series is convergent: $|\sum a_n| \le \sum |a_n|$.

Section 4: Alternating Series Error Bound

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10 For a convergent alternating series $\sum(-1)^{n-1}b_n$, what is the upper bound for the error $|R_n| = |S - S_n|$?

a. $|R_n| \le b_{n+1}$ ✓

b. $|R_n| \le b_n$

c. $|R_n| \le \frac{b_{n+1}}{2}$

d. $|R_n| \le \frac{b_n}{n}$

The Alternating Series Estimation Theorem states that the truncation error $|S - S_n|$ is bounded by the magnitude of the first omitted term: b_{n+1}.

11 If the sum of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3}$ is approximated by its first 4 terms, what is the maximum possible error?

a. $\frac{1}{125}$ ✓

b. $\frac{1}{64}$

c. $\frac{1}{216}$

d. $\frac{1}{25}$

The first omitted term is the 5th term: $b_5 = \frac{1}{5^3} = \frac{1}{125} = 0.008$.

12 How many terms of the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ are needed to guarantee an error of less than 0.01?

a. 100 **terms** ✓

b. 99 terms

c. 10 terms

d. 50 terms

We require $b_{n+1} = \frac{1}{n+1} < 0.01 = \frac{1}{100} \Rightarrow n + 1 > 100 \Rightarrow n \ge 100$ terms.

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